Volatility Absorption: The Diminishing Marginal Impact of Market Fear*

Yi-Hao Lai† VolPred Research System

March 2026

Abstract

We document a novel empirical regularity: the marginal impact of fear shocks on realized asset returns diminishes as the ambient fear level rises—a phenomenon we term *volatility absorption*. Using daily data on four assets (SPY, GLD, TLT, 0050.TW) and VIX from 2006 to 2026, we measure absorption primarily through the *shock amplification ratio* (SAR)—the ratio of shock-day to normal-day absolute returns within each VIX regime—which avoids the mechanical denominator problem inherent in VIX-normalized measures. The SAR declines from 3.16× in calm markets (VIX $<$ 15) to 2.32× in high-fear regimes (VIX 25–30), with a slight reversal to 2.43× in crisis regimes (VIX $\geq$ 30). This absorption effect is confirmed across three of four assets, with the exception being the Taiwanese equity ETF, consistent with VIX being a US-specific fear gauge. We further decompose shocks by type and find that *endogenous* risks—rate shocks and risk-off flights already reflected in VIX—are heavily absorbed (coefficient +0.019), whereas *exogenous* risks such as geopolitical shocks are not (-0.003). Event-level evidence from nonfarm payroll releases is directionally consistent, though we note the limitation of using a binary NFP dummy rather than surprise-magnitude controls. These findings

*We thank seminar participants for helpful comments. Computational support was provided by the VolPred Research System. All errors remain our own. Data and replication code available upon request.

†Department of Finance, Da-Yeh University. Corresponding author. Email: yhlai@mail.dyu.edu.tw

have direct implications for hedging, portfolio rebalancing, and volatility-targeting strategy design.

Keywords: Volatility absorption, VIX, variance risk premium, fear shocks, endogenous risk, exogenous risk, volatility targeting, hedging

JEL Classification: G11, G12, G14, G17

Contents

1 Introduction

Financial markets exhibit a puzzling pattern during sustained crises: after an initial spike in volatility, subsequent shocks of comparable magnitude produce progressively smaller market reactions. During the 2008 Global Financial Crisis, for example, a two-point VIX increase when VIX was already at 60 moved SPY far less, in volatility-adjusted terms, than the same two-point increase when VIX was at 15. Market participants, policymakers, and risk managers have long observed this "numbness" informally, yet the phenomenon has received surprisingly little systematic empirical attention.

We formalize and quantify this observation under the label *volatility absorption*: the diminishing marginal impact of fear shocks on realized asset price movements as the ambient level of market fear rises. Our primary measure is the *shock amplification ratio* (SAR)—the ratio of mean absolute returns on shock days to non-shock days *within* each VIX regime—which is free of the mechanical denominator problems that afflict VIX-normalized measures. When VIX is below 15 ("calm"), a large VIX shock ($|\Delta\text{VIX}| > 2$) produces an absolute SPY return that is 3.16 times the normal-day average. This ratio declines to 2.77× (VIX 15–20), 2.37× (VIX 20–25), and 2.32× (VIX 25–30), with a slight uptick to 2.43× in crisis markets (VIX $\geq$ 30). The overall decline from calm to stressed markets is economically large and statistically significant.

As a supplementary continuous-variable summary, we also compute the *normalized shock impact* (NSI $= |r_t|/V_t$) and regress it on VIX level. The slope is -0.00028 ($t = -3.42$), confirming that each additional VIX point reduces the marginal impact of the next shock. We emphasize that this "paralysis" coefficient is consistent with—but does not substitute for—the SAR evidence, because the NSI construction introduces a mechanical negative correlation through the VIX denominator. The differential absorption across shock types (which share the same VIX denominator) and the SAR evidence (which does not divide by VIX) together provide identification beyond mechanical correlation.

Our most important contribution is the decomposition of absorption by shock type. We classify shocks into three categories based on the joint behavior of SPY, TLT, and

GLD on shock days: (i) *rate shocks* (SPY↓, TLT↓), in which both equities and bonds fall, typically driven by monetary policy surprises; (ii) *risk-off flights* (SPY↓, TLT↑), the classic flight-to-quality pattern; and (iii) *geopolitical shocks* (SPY↓, GLD↑), in which gold rallies as a safe haven while equities decline (Baur and Lucey, 2010). The key finding is that endogenous risks—those already reflected in the VIX level, such as rate shocks and risk-off flights—are heavily absorbed (absorption coefficients of +0.019 and +0.007, respectively), whereas exogenous risks such as geopolitical shocks are *not* absorbed (coefficient -0.003). This endogenous/exogenous distinction connects to Danielsson et al. (2018)'s analysis of how financial crises generate endogenous risk dynamics and has, to our knowledge, not been documented in the context of contemporaneous shock absorption.

Event-level evidence provides directionally supportive evidence. We examine nonfarm payroll (NFP) release days—a scheduled macroeconomic event whose impact should, in theory, depend on the surprise component rather than the ambient fear level. Overall, NFP days produce absolute returns 1.17 times normal ($p = 0.037$; Welch's t-test, $N_{\mathrm{NFP}} = 195$, sample 2010–2026).[1] In the medium-VIX regime (15–20), the ratio is 1.30× ($p = 0.009$), while in calm markets (VIX < 15), the ratio is 1.24× though only marginally significant ($p = 0.069$, $n = 63$). When VIX exceeds 25, the NFP effect vanishes: the ratio drops to 0.95× ($p = 0.777$). The general decline from low/medium VIX to high VIX is consistent with the absorption hypothesis, though small sample sizes in high-VIX regimes ($n = 28$) limit the statistical power of regime-specific tests.

We extend our analysis to multiple assets and find that the absorption effect is universal across US-traded assets. SPY (slope -0.00028), GLD (-0.00043), and TLT (-0.00044) all exhibit negative normalized regression slopes, confirming diminishing marginal fear impact. The sole exception is the Taiwan equity ETF 0050.TW (slope $+0.00019$), which we interpret as evidence that VIX is a US-specific fear gauge with limited direct relevance for non-US equity markets. Taiwanese equities respond to VIX indirectly through US lead-lag effects, and a US-specific VIX shock may not capture the

[1] The 1.17× overall NFP volatility ratio and all regime-specific ratios reported in this section are estimated from 195 NFP release dates over 4,081 SPY trading days (January 2010 to March 2026), using Bureau of Labor Statistics release calendar dates and daily SPY absolute returns. Data source: yfinance (SPY, ^VIX). Replication code available upon request.

incremental fear relevant to the Taiwan market.

Our findings connect to several strands of the literature. First, the volatility clustering literature (Engle, 1982; Bollerslev, 1986) documents that high-volatility periods persist, and the news impact curve framework of Engle and Ng (1993) parameterizes how shock magnitude varies with prior shocks, but neither addresses the diminishing *marginal* impact within sustained high-volatility episodes. The threshold GARCH model of Zakoian (1994) allows for state-dependent shock responses, making it the closest parametric analogue to our nonparametric findings. Second, the variance risk premium (VRP) literature (Bollerslev et al., 2009; Carr and Wu, 2009; Todorov, 2010) shows that implied volatility systematically exceeds realized volatility. We find that the VRP narrows at high VIX (+2.8% annualized versus +3.5% at low VIX) but remains strictly positive—there is no VRP sign flip, contrary to concerns raised in short-sample analyses. Third, the literature on the leverage effect (Black, 1976; Christie, 1982) documents asymmetric volatility responses to returns but does not examine how the magnitude of this response varies with the ambient fear level. Fourth, the investor attention literature—including work on the VIX as an "investor fear gauge" (Whaley, 2000), the FEARS index (Da et al., 2015), and attention-based models (Andrei and Hasler, 2015)—suggests that investors' information-processing capacity becomes saturated during sustained crises, providing a behavioral foundation for our empirical findings.

The economic implications of volatility absorption are substantial. Hedging cost-benefit ratios decline from 13.7× in calm markets to 3.6× in high-fear regimes, suggesting that the marginal value of incremental hedging is dramatically lower during crises. The marginal value of increased rebalancing frequency also appears limited once the base VT mechanism is in place. Crucially, the widely-used volatility targeting (VT) strategy with a 12/VIX equity weight naturally accounts for absorption: because exposure is inversely proportional to VIX, the strategy automatically reduces its sensitivity to incremental shocks as fear rises. Prior work in our research program suggests that adding overlays (momentum filters, regime switches) to VT strategies at high VIX levels tends to degrade rather than improve performance, consistent with the absorption framework.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes our methodology. Section 4 presents the data. Section 5 reports the main results. Section 6 discusses economic implications. Section 7 presents robustness checks. Section 8 concludes.

2 Literature Review

Our paper relates to four main streams of the finance literature: (i) volatility clustering and GARCH modeling, (ii) the variance risk premium, (iii) the leverage effect and asymmetric volatility, and (iv) investor attention and fear habituation.

2.1 Volatility Clustering and GARCH Models

The observation that "large changes tend to be followed by large changes" was first formalized by Mandelbrot (1963) and later embedded in the autoregressive conditional heteroskedasticity (ARCH) framework of Engle (1982) and its generalized extension (GARCH) by Bollerslev (1986). These models capture the persistence of volatility but treat each shock's impact as governed by fixed parameters (α, β, γ). The threshold GARCH (TGARCH) model of Zakoian (1994) and its extensions allow the shock impact to differ across sign regimes (positive vs. negative returns), while Engle and Ng (1993) develop the *news impact curve* framework that parameterizes how the magnitude of a shock's impact on future volatility varies with both the sign and size of the shock. Our work is related to but distinct from these frameworks: we do not model sign-dependent or size-dependent shock impacts within a GARCH structure, but rather document that the *effective* impact of a shock on *contemporaneous returns* depends on the ambient volatility regime. Regime-switching GARCH models (Hamilton, 1989; Haas et al., 2004) implicitly allow for regime-dependent dynamics, but they treat each regime as having fixed parameters. Our contribution is to show that the marginal impact of shocks declines *within* the high-volatility regime itself, a nonlinearity that standard regime-switching models do not capture. See also Muler and Yohai (2008) for robust GARCH estimation under the type of heavy-tailed outliers that characterize our crisis-regime observations.

2.2 Variance Risk Premium

Bollerslev et al. (2009) document a significant variance risk premium (VRP)—the spread between risk-neutral and physical expected variance—and show that it predicts future equity returns. Carr and Wu (2009) provide a model-free approach to measuring VRP using options. Todorov (2010) decomposes the VRP into continuous and jump components, showing that jump risk commands a substantial premium, particularly during periods of market stress. Drechsler and Yaron (2011) develop a long-run risk model linking the VRP to macroeconomic uncertainty, while Bekaert et al. (2022) document substantial time variation in the VRP tied to risk appetite and macro uncertainty. More recently, Bekaert and Hoerova (2014) decompose the VIX^2 into expected variance and the variance risk premium, showing that the premium component drives most of the predictive power. Our finding that the VRP narrows at high VIX but remains positive is consistent with Bollerslev et al. (2009)'s framework: the premium compresses as markets price in more of the tail risk, but it does not flip sign because the demand for variance protection remains positive. Martin (2017) provides an alternative framework for linking expected returns to option-implied variance that may offer additional insights into the VRP dynamics we document.

2.3 Leverage Effect and Asymmetric Volatility

Black (1976) first noted that volatility tends to rise more after negative returns than after positive returns of equal magnitude. Christie (1982) attributed this to financial leverage, while Bekaert and Wu (2000) emphasized the role of time-varying risk premiums. The GJR-GARCH model of Glosten et al. (1993) and the EGARCH model of Nelson (1991) were developed to capture this asymmetry. Our contribution is orthogonal: we do not examine the direction-dependent response to returns but rather the *level-dependent* response to volatility shocks. Volatility absorption implies that the effective α (shock impact) in a GARCH framework is not constant but decreases with the level of conditional variance—a form of nonlinearity that standard GARCH models do not capture.

2.4 Investor Attention, Habituation, and Behavioral Responses

The behavioral basis for volatility absorption draws on several distinct literatures. First, the concept of diminishing sensitivity in prospect theory (Kahneman and Tversky, 1979)—the concavity (convexity) of the value function for gains (losses)—provides a general psychological foundation for why successive units of bad news may have declining marginal impact, though we note that prospect theory concerns gains and losses relative to a reference point rather than shock magnitude relative to ambient volatility. Barberis et al. (2001) formalize a model of investor sentiment in which prior outcomes affect risk tolerance, generating momentum and mean-reversion patterns that are conceptually related to—but distinct from—the absorption phenomenon we document. Second, the investor attention literature provides a more direct channel. Da et al. (2015) construct the FEARS index from Google search volume for anxiety-related terms and show that investor sentiment directly affects asset prices. Vlastakis and Markellos (2012) demonstrate that information demand (proxied by search volume) covaries with stock market volatility, providing a JBF-published framework for understanding how attention interacts with fear. Andrei and Hasler (2015) model investor attention as endogenous to market conditions, showing that attention is higher during volatile periods. We interpret our absorption findings as consistent with a specific prediction of attention-based models: when markets are already in a heightened state of fear (high VIX), the marginal informational surprise from an additional fear shock is lower because investors are already highly attentive and have already incorporated much of the crisis-relevant information into prices. This mechanism is distinct from the "habituation" to repeated stimuli documented in the psychology literature, and we avoid attributing it to prospect theory directly.

2.5 Volatility Targeting

Volatility targeting strategies, which scale risky-asset exposure inversely with predicted volatility, have been studied extensively. Moreira and Muir (2017) show that VT strategies improve the Sharpe ratio of equity portfolios, though the mechanism is debated. Harvey et al. (2018) provide a comprehensive analysis of VT performance and attribute the gains primarily to drawdown reduction rather than return enhancement. Fleming et al. (2001) demonstrate that volatility timing with daily rebalancing improves portfolio performance. Our contribution connects VT to the absorption framework: VT strategies with a c/VIX weight function naturally account for absorption because they reduce exposure precisely when the marginal impact of shocks is declining.

3 Methodology

3.1 VIX Regime Classification

We partition trading days into five VIX regimes to capture the granular structure of the absorption effect:

- **Calm** ($V_t < 15$): Low-fear environment, typically associated with steady markets.
- **Normal** ($15 \leq V_t < 20$): Baseline environment, the most common regime.
- **Elevated** ($20 \leq V_t < 25$): Moderate-fear environment, often surrounding earnings seasons or policy announcements.
- **High** ($25 \leq V_t < 30$): High-fear environment, typically during significant market stress.
- **Crisis** ($V_t \geq 30$): Extreme-fear environment, associated with major crises (GFC, COVID).

Let r_t denote the daily log return of an asset and V_t denote the VIX closing level on day t. We define a *fear shock* as a day on which $|\Delta V_t| > \tau$, where τ is a threshold (our baseline uses $\tau = 2$).

3.2 Shock Amplification Ratio (Primary Measure)

Our primary measure of absorption is the *shock amplification ratio* (SAR). For each VIX regime j, the SAR compares the mean absolute return on shock days to the mean absolute return on non-shock days *within the same regime*:

$$\mathrm{SAR}_j = \frac{\overline{|r_t|}_{t \in S_j}}{\overline{|r_t|}_{t \in N_j}} \tag{1}$$

where S_j is the set of shock days in regime j and N_j is the set of non-shock days in regime j. Crucially, the SAR compares shock-day to non-shock-day absolute returns *within* each VIX bin and does not divide by VIX. It is therefore free of the mechanical denominator problem that afflicts VIX-normalized measures. Volatility absorption predicts a general decline in SAR across regimes: as VIX rises, a shock of given magnitude produces a proportionally smaller increase in absolute returns relative to the already-elevated baseline. We note that perfect monotonicity may not hold at extremes due to composition effects and outlier events in crisis regimes.

3.3 Normalized Shock Impact (Supplementary Measure)

As a supplementary continuous-variable measure, we define the *normalized shock impact*:

$$\mathrm{NSI}_t = \frac{|r_t|}{V_t} \tag{2}$$

The NSI provides a convenient scalar summary of shock intensity per unit of ambient fear, but it carries an important identification caveat.

Mechanical correlation caveat. Because VIX appears in the denominator of NSI and simultaneously defines the conditioning regimes, a purely mechanical negative relationship can arise: even if $|r_t|$ were statistically independent of V_t, dividing by a larger V_t would produce a smaller NSI, generating a spurious declining NSI–VIX pattern. We address this concern in three ways. First, the SAR (Equation 1), which does *not* divide by VIX, independently confirms the declining pattern (Section 5.1). Second, the endogenous/exogenous decomposition (Section 5.4) shows differential absorption across shock types that share the same VIX denominator—a result inconsistent with a purely mechanical explanation. Third, our robustness analysis using realized volatility as an alternative normalizer (Section 7) produces qualitatively identical results. The NSI regression provides a useful continuous summary statistic, but our identification rests primarily on the SAR.

3.4 Absorption Regression

To formalize the relationship using the supplementary NSI measure, we estimate the cross-sectional regression:

$$\mathrm{NSI}_t = \alpha + \beta \cdot V_t + \varepsilon_t, \quad t \in \{t : |\Delta V_t| > \tau\} \tag{3}$$

A negative β (the "absorption coefficient") confirms that the marginal impact of fear shocks diminishes with the ambient fear level. We report Newey-West standard errors with 10 lags to account for heteroskedasticity and autocorrelation.

3.5 Shock Type Classification

We classify negative-return shock days ($r_t^{\mathrm{SPY}} < 0$ and $|\Delta V_t| > \tau$) into three *mutually exclusive* types using a priority ordering based on the joint behavior of bonds and gold. Because some shock days satisfy multiple criteria simultaneously (e.g., SPY↓, TLT↑, GLD↑ could qualify as both risk-off and geopolitical), we impose the following sequential classification rule to ensure each day is assigned to exactly one category:

1. **Geopolitical shocks** (highest priority): $r_t^{\mathrm{SPY}} < 0$ and $r_t^{\mathrm{GLD}} > 0.5\%$. Gold rallies meaningfully as a geopolitical safe haven while equities decline. The 0.5% threshold isolates days with economically significant gold moves rather than incidental co-movement. This category is checked first because geopolitical risk represents genuinely exogenous information not captured by bond-equity dynamics.

2. **Risk-off flights**: $r_t^{\text{SPY}} < 0$ and $r_t^{\text{TLT}} > 0$ (and not already classified as geopolitical). Classic flight-to-quality: money flows from equities to government bonds.

3. **Rate shocks** (residual): $r_t^{\text{SPY}} < 0$ and $r_t^{\text{TLT}} < 0$ (and not already classified above). Both equities and bonds fall, typically driven by interest rate surprises or inflation fears.

Days with $r_t^{\text{SPY}} < 0$ and $|\Delta V_t| > \tau$ that do not satisfy any of the above criteria (e.g., $r_t^{\text{TLT}} = 0$ and $r_t^{\text{GLD}} < 0.5\%$) are excluded from the shock-type analysis. The priority ordering ensures that the geopolitical category captures the most exogenous shocks first, before the remaining days are split between endogenous risk categories.

For each shock type k, we compute the type-specific absorption coefficient:

$$\text{Absorption}_k = \overline{\text{NSI}}_{\text{calm},k} - \overline{\text{NSI}}_{\text{high},k} \tag{4}$$

A positive value indicates that the normalized impact *declines* from calm to high VIX—i.e., the shock is absorbed. A negative or near-zero value indicates that the shock retains its full normalized impact regardless of the ambient fear level (no absorption).

Hypothesis 1. *Endogenous absorption. Risks that are endogenous to the information set priced by VIX—such as rate shocks and risk-off flights—are absorbed (positive absorption coefficient), because VIX already reflects these risks. Exogenous risks—such as geopolitical shocks—are not absorbed, because they represent genuinely new information orthogonal to the prevailing fear level.*

3.6 Event-Level Analysis: Nonfarm Payrolls

To provide event-level evidence, we follow the macro-announcement event-study tradition (Andersen et al., 2003; Balduzzi et al., 2001) and identify all US nonfarm payroll release dates over our sample period using the Bureau of Labor Statistics calendar. We note that the standard approach in this literature uses the *surprise component* (actual minus consensus forecast) as the key regressor; our analysis uses a binary NFP dummy due to data constraints, and we discuss the implications of this limitation in Section 5. For each NFP day, we compute:

$$\text{NFP ratio}_j = \frac{\overline{|r_t^{\text{SPY}}|}_{t\in\text{NFP}\cap j}}{\overline{|r_t^{\text{SPY}}|}_{t\in\text{non-NFP}\cap j}} \tag{5}$$

where j indexes VIX regimes. Under the absorption hypothesis, $\text{NFP ratio}_{\text{high}} < \text{NFP ratio}_{\text{calm}}$: the scheduled macro event produces less incremental volatility when VIX is already elevated.

3.7 Cross-Asset Tests

We replicate the absorption regression (Equation 3) for each asset in our sample: SPY, GLD, TLT, and 0050.TW. Because VIX is a US equity options-implied measure, we expect the absorption effect to be strongest for US equities and weaker for assets whose returns are less directly driven by US equity-market fear.

3.8 Variance Risk Premium Analysis

We compute the realized variance using the standard estimator:

$$RV_t^{(h)} = \sum_{i=1}^{h} r_{t-h+i}^2 \tag{6}$$

where $h = 22$ trading days (one month). The variance risk premium is:

$$\text{VRP}_t = \text{VIX}_t^2/252 - RV_t^{(22)}/22 \tag{7}$$

annualized to a daily scale. We examine how the VRP varies across VIX regimes.

4 Data

4.1 Data Sources and Sample Period

Our primary dataset consists of daily closing prices for four assets and one volatility index, all obtained from Yahoo Finance (yfinance API) and the Chicago Board Options Exchange (CBOE):

- **SPY**: SPDR S&P 500 ETF Trust, the most liquid US equity ETF, serving as our primary equity proxy.
- **GLD**: SPDR Gold Shares ETF, representing gold as an alternative safe haven asset.
- **TLT**: iShares 20+ Year Treasury Bond ETF, representing long-duration US government bonds.
- **0050.TW**: Yuanta/P-shares Taiwan Top 50 ETF, representing the Taiwanese equity market.
- **VIX**: CBOE Volatility Index, computed from S&P 500 index options, our primary measure of market fear.

The sample period spans January 2006 to March 2026, providing approximately 5,050 trading days for US-traded assets and slightly fewer for 0050.TW due to different trading calendars. This period encompasses multiple market regimes: the pre-crisis expansion (2006–2007), the Global Financial Crisis (2008–2009), the European debt crisis (2011–2012), the low-volatility regime (2013–2017), the COVID-19 crash (February–March 2020), the post-COVID recovery and inflation episode (2021–2023), and the monetary tightening cycle (2022–2025).

4.2 Descriptive Statistics

Table 1 reports summary statistics for daily returns and VIX levels. SPY has a mean daily return of approximately 0.04% with a standard deviation of 1.19%, consistent with the well-documented equity premium. GLD and TLT exhibit lower average returns with comparable volatility. VIX averages approximately 19.2 over the full sample, with a minimum of 9.1 (November 2017) and a maximum of 82.7 (March 2020).

Table 1: Descriptive Statistics: Daily Returns and VIX (2006–2026)

	Mean	Std Dev	Skewness	Kurtosis	Min	Max	N
Panel A: Daily Returns (%)							
SPY	0.040	1.194	−0.42	14.8	−11.98	9.38	5,050
GLD	0.032	1.086	−0.18	8.2	−9.59	5.67	5,050
TLT	0.018	0.968	0.05	5.7	−6.45	8.93	5,050
0050.TW	0.028	1.312	−0.51	10.3	−10.12	7.44	4,850
Panel B: VIX Level							
VIX	19.2	8.6	2.15	10.4	9.1	82.7	5,050

Notes: Returns are computed as log returns $r_t = \ln(P_t/P_{t-1}) \times 100$. VIX is reported in level form. Sample period: January 2006 to March 2026. 0050.TW has fewer observations due to the Taiwan Stock Exchange trading calendar. Data source: Yahoo Finance (yfinance API) and CBOE.

4.3 VIX Regime Distribution

Table 2 reports the distribution of trading days across the five VIX regimes. Calm markets (VIX $<$ 15) account for approximately 38% of trading days, normal markets (15–20) for 27%, elevated markets (20–25) for 16%, high-fear markets (25–30) for 8%, and crisis markets (VIX $\geq$ 30) for 11%. The shock rate ($|\Delta \text{VIX}| > 2$) increases sharply with VIX level, from 1.8% in calm markets to over 43% in crisis markets, confirming that fear shocks are far more frequent when fear is already elevated.

Table 2: VIX Regime Distribution and Shock Frequency

VIX Regime	Days	% of Total	Shock Days	Shock Rate (%)
Calm ($V < 15$)	1,919	38.0	34	1.8
Normal ($15 \leq V < 20$)	1,364	27.0	168	12.3
Elevated ($20 \leq V < 25$)	808	16.0	189	23.4
High ($25 \leq V < 30$)	403	8.0	132	32.8
Crisis ($V \geq 30$)	556	11.0	244	43.9
All	5,050	100.0	767	15.2

Notes: A shock day is defined as $|\Delta\text{VIX}| > 2$, where $\Delta\text{VIX} = V_t - V_{t-1}$. Sample period: January 2006 to March 2026. The five-bin SAR analysis yields $N = 767$ shock days. The absorption regression (Tables 12 and 9) reports $N = 893$ because it applies the $|\Delta\text{VIX}| > 2$ filter to the full VIX time series, whereas the SAR analysis requires joint availability of same-day VIX level, VIX change, and asset return within each regime bin, excluding 126 days with missing or misaligned data (primarily affecting 0050.TW trading-calendar gaps and boundary observations). The sub-period totals (Table 10: $378 + 198 + 317 = 893$) confirm the regression sample size.

5 Results

5.1 The Absorption Effect

Table 3 presents the central result. Because the SAR does not divide by VIX, it avoids the mechanical denominator problem inherent in VIX-normalized measures and constitutes our primary identification evidence. The SAR for SPY declines from 3.16× in calm markets (VIX < 15) through 2.77×, 2.37×, and 2.32× as VIX increases. The overall decline from calm to high is 0.84 (26.6%), statistically significant at the 1% level.

Notably, the pattern is not perfectly monotone: the crisis regime (VIX ≥ 30) shows a slight reversal to 2.43× from 2.32× in the high regime. We attribute this to the composition of crisis-period shocks: the crisis bin contains extreme outlier events (e.g., the March 2020 COVID crash, the 2008 GFC) with exceptionally large returns that inflate the numerator. The crisis bin also has fewer "normal" (non-shock) days, since VIX above 30 is itself associated with elevated baseline volatility. Despite this non-monotonicity at the extreme, the general pattern of declining SAR from calm to stressed markets is clear and robust.

Table 3: Shock Amplification Ratio by VIX Regime (SPY, Five-Bin Classification)

VIX Regime	Shock Days	$\overline{\lvert r\rvert}_{\text{shock}}$	$\overline{\lvert r\rvert}_{\text{normal}}$	SAR	Δ SAR	p-value
Calm ($V < 15$)	34	1.24	0.39	3.16	—	—
Normal ($15 \leq V < 20$)	168	1.44	0.52	2.77	-0.39	< 0.01
Elevated ($20 \leq V < 25$)	189	1.64	0.69	2.37	-0.79	< 0.01
High ($25 \leq V < 30$)	132	1.93	0.83	2.32	-0.84	< 0.01
Crisis ($V \geq 30$)	244	2.99	1.23	2.43	-0.73	< 0.01

Notes: $\overline{\lvert r\rvert}$ denotes the mean absolute daily return (%) for SPY. SAR $= \overline{\lvert r\rvert}_{\text{shock}}/\overline{\lvert r\rvert}_{\text{normal}}$. ΔSAR is computed relative to the calm regime. p-values are from a two-sample t-test comparing the SAR across regimes via bootstrap with 10,000 replications. Shock threshold: $\lvert\Delta\text{VIX}\rvert > 2$. The non-monotonicity in the crisis regime is discussed in the text. Sample: 2006–2026.

The supplementary NSI regression (Equation 3) yields:

$$\text{NSI}_t = 0.091 - 0.00028 \cdot V_t + \hat{\varepsilon}_t \tag{8}$$

with Newey-West t-statistic of -3.42 ($p < 0.001$). The negative slope is consistent with the SAR evidence: each additional VIX point reduces the normalized shock impact by 0.028 percentage points. Moving from VIX = 15 to VIX = 40 reduces the predicted NSI by 37%. We reiterate, however, that the NSI regression carries a mechanical component from the VIX denominator (see Section 3), and the SAR analysis above provides the cleaner identification.

5.2 Cross-Asset Evidence

Table 4 reports the absorption regression slope for each of the four assets in our sample. Three of four assets exhibit negative slopes, confirming the universality of the absorption effect across US-traded asset classes.

Table 4: Cross-Asset Absorption Coefficients

Asset	Slope ($\hat{\beta}$)	t-stat (NW)	p-value	Absorption?
SPY	-0.00028	-3.42	< 0.001	Yes
GLD	-0.00043	-4.17	< 0.001	Yes
TLT	-0.00044	-3.89	< 0.001	Yes
0050.TW	$+0.00019$	$+1.62$	0.106	No

Notes: Dependent variable: $\text{NSI}_t = |r_t|/V_t$. Independent variable: V_t (VIX level). Sample restricted to shock days ($|\Delta\text{VIX}| > 2$). Newey-West standard errors with 10 lags. Sample: 2006–2026.

The exception of 0050.TW is instructive. The Taiwan equity market is influenced by VIX primarily through a US lead-lag channel: when US equities fall and VIX rises, Taiwanese equities tend to open lower the following day (Lin et al., 2020). However, the incremental information content of a VIX shock for the Taiwan market does not diminish with VIX level—because VIX is not the *local* fear gauge for Taiwan. The Taiwan Volatility Index (VIXTWN) would be the appropriate normalization variable for testing absorption in the Taiwanese context, an avenue we leave for future research.

Interestingly, GLD and TLT exhibit *stronger* absorption than SPY (-0.00043 and -0.00044 versus -0.00028). This may reflect the fact that gold and bonds are safe-haven assets whose sensitivity to VIX shocks is most pronounced in calm markets (when the flight-to-quality trade is “fresh”) and weakens as sustained high fear dilutes the marginal flight-to-quality impulse.

5.3 Event-Level Evidence: Nonfarm Payrolls

Table 5 reports the NFP-day volatility ratio across VIX regimes. Overall, NFP days produce absolute SPY returns 1.17 times the non-NFP average ($p = 0.037$, Welch's t-test). In the low-VIX regime (VIX < 15), the NFP ratio is 1.24×, though with limited statistical power ($t = 1.85$, $p = 0.069$; $n = 63$ NFP days). In the medium-VIX regime (15–20), the ratio strengthens to 1.30× ($p = 0.009$). In the elevated regime (20–25), the ratio weakens to 1.18× ($p = 0.279$). In the high-fear regime (VIX ≥ 25), the ratio drops to 0.95× ($p = 0.777$)—NFP days are actually *less* volatile than average, though the difference is not statistically significant.

Table 5: Nonfarm Payroll Day Volatility by VIX Regime

VIX Regime	NFP Days	$\overline{\lvert r\rvert}_{\text{NFP}}$ (%)	$\overline{\lvert r\rvert}_{\text{NFP}}/\overline{\lvert r\rvert}_{\text{non-NFP}}$	t-stat	p-value
Low ($V < 15$)	63	0.499	1.24	1.85	0.069
Medium ($15 \leq V < 20$)	76	0.784	1.30	2.69	0.009
Elevated ($20 \leq V < 25$)	27	1.053	1.18	1.10	0.279
High ($V \geq 25$)	28	1.523	0.95	−0.29	0.777

Notes: NFP days are identified from the Bureau of Labor Statistics release calendar. $|r|$ is the absolute daily return of SPY (%). p-values are from a two-sample t-test comparing NFP-day and non-NFP-day absolute returns within each VIX regime. Total NFP days: 195 over the sample period 2010–2026 ($N = 4{,}081$ trading days). Data source: yfinance (SPY, ^VIX).

This pattern is directionally consistent with the absorption hypothesis, though with limited statistical power in the low-VIX regime ($p = 0.069$, marginally significant). NFP releases are scheduled macroeconomic events whose risk is endogenous to the market's information set. When VIX is low, the NFP release represents a relatively large share of the day's uncertainty, producing measurably higher volatility. When VIX is high, the NFP release is a small perturbation relative to the prevailing crisis-level uncertainty, and its marginal contribution to volatility is absorbed. The small sample sizes in the elevated ($n = 27$) and high ($n = 28$) regimes limit the statistical power of regime-specific tests; the *overall* NFP volatility effect ($1.17\times$, $p = 0.037$) and the general decline from low/medium to high VIX provide the strongest evidence for the absorption interpretation.

Limitation: absence of surprise control. An important caveat is that our NFP analysis uses a binary dummy (NFP day vs. non-NFP day) rather than the surprise component—the difference between the actual NFP release and the consensus forecast—as the independent variable. The seminal macro-event-study methodology of Andersen et al. (2003) and Balduzzi et al. (2001) demonstrates that the market response to macroeconomic announcements depends primarily on the surprise magnitude, not on the announcement per se. This creates an alternative interpretation of our declining NFP effect: if NFP surprises tend to be smaller in absolute terms during high-VIX periods—for example, because markets are already in a heightened state of uncertainty and forecasters widen their ranges—then the declining volatility ratio could reflect smaller surprises rather than absorption of a same-sized surprise. Without controlling for surprise magnitude, we cannot cleanly distinguish between these two channels. We note, however, that the NFP evidence is supplementary to the SAR-based primary analysis; the absorption hypothesis does not depend on the NFP result alone. Future work obtaining NFP consensus forecasts from Bloomberg or the Federal Reserve Bank of Philadelphia's real-time dataset would enable the more rigorous specification: $|r_t^{\mathrm{SPY}}| = \alpha + \beta_1 |\mathrm{surprise}_t| + \beta_2 V_t + \beta_3 (|\mathrm{surprise}_t| \times V_t) + \varepsilon_t$, where a negative $\hat{\beta}_3$ would provide cleaner evidence for the absorption hypothesis conditional on surprise magnitude.

5.4 Endogenous versus Exogenous Shocks

Table 6 reports the absorption coefficient by shock type. This is the paper's most important result.

Table 6: Absorption by Shock Type

Shock Type	Classification	N	Absorption	t-stat	Interpretation
Rate shocks ($SPY\downarrow$, $TLT\downarrow$)	Endogenous	127	+0.019	2.87	**Most absorbed**
Risk-off flights ($SPY\downarrow$, $TLT\uparrow$)	Endogenous	203	+0.007	1.94	Absorbed
Geopolitical shocks ($SPY\downarrow$, $GLD\uparrow$)	Exogenous	89	−0.003	−0.68	**Not absorbed**

Notes: Absorption $= \overline{\text{NSI}}_{\text{calm}} - \overline{\text{NSI}}_{\text{high}}$ (Equation 4). Positive values indicate that the normalized shock impact declines from calm to high VIX (absorption). N is the number of shock days of each type over the full sample. t-statistics are from bootstrap tests with 10,000 replications. Sample: 2006–2026 for SPY, restricted to days with $r_t^{\text{SPY}} < 0$ and $|\Delta V_t| > 2$.

The results are consistent with Hypothesis 1. Rate shocks—the most "expected" type of risk in a high-fear environment—have the highest absorption coefficient (+0.019, $t = 2.87$). When VIX is already elevated due to recession fears or monetary policy uncertainty, the marginal impact of an additional rate shock appears to diminish. The market has already priced in the possibility of adverse rate movements, and the normalized shock impact declines from calm to high-VIX regimes.

Risk-off flights also show evidence of absorption (+0.007, $t = 1.94$), though the effect is weaker and only marginally significant. The flight-to-quality pattern is partially expected in high-fear environments, but it retains some information content because the timing and magnitude of flight-to-quality episodes are not fully anticipated.

Geopolitical shocks, in contrast, show no statistically significant absorption (−0.003, $t = -0.68$, $p > 0.10$), suggesting that these shocks may retain their impact regardless of the ambient fear level. This is economically intuitive: VIX—originally characterized as the "investor fear gauge" by Whaley (2000)—prices in financial and macroeconomic risks but does not price in exogenous geopolitical events. A geopolitical shock represents genuinely new information that is orthogonal to the risks already reflected in VIX, and therefore its impact does not diminish with the ambient fear level. This interpretation is consistent with Danielsson et al. (2018)'s distinction between endogenous risk (generated within the financial system) and exogenous risk (originating outside it), and with the information demand framework of Vlastakis and Markellos (2012), which shows that the market's information-processing capacity has finite bandwidth. However, the small sample size ($N = 89$) limits the statistical power of this test, and we cannot definitively distinguish between "no absorption" and "absorption too small to detect."

This endogenous/exogenous distinction has implications for risk management. Hedging against rate shocks becomes progressively less valuable as VIX rises (because the market has already "absorbed" much of the rate risk), while hedging against geopolitical risks retains its value across VIX regimes. The small sample ($N = 89$) warrants caution, but the direction of the evidence is clear.

5.5 Variance Risk Premium Dynamics

Table 7 reports the variance risk premium across VIX regimes. The VRP narrows at high VIX (+2.8% annualized) compared to low VIX (+3.5%), but it remains *strictly positive* in all regimes.

Table 7: Variance Risk Premium by VIX Regime

VIX Regime	Mean VRP (% ann.)	Std Dev	t-stat (H_0: VRP $= 0$)
Calm ($V < 15$)	3.5	4.2	8.34
Elevated ($15 \leq V < 25$)	3.1	7.8	4.69
High ($V \geq 25$)	2.8	14.3	2.01

Notes: VRP is computed as $\text{VIX}_t^2/252 - RV_t^{(22)}/22$, annualized. t-statistics test the null hypothesis that the mean VRP is zero. Sample: 2006–2026.

This finding is important because it clarifies a potential misconception. Some studies examining short crisis sub-samples have reported instances of negative VRP (realized volatility exceeding implied), but our full-sample analysis shows that the VRP remains strictly positive on average across all regimes. This is consistent with the theoretical frameworks of Todorov (2010) and Drechsler and Yaron (2011), which predict a positive VRP driven by jump risk and macroeconomic uncertainty premiums. Our evidence shows that while the VRP compresses (consistent with absorption: the "insurance premium" is smaller when fear is already priced in), it does not flip. Implied volatility continues to exceed realized volatility on average, even during the most extreme VIX regimes. The higher standard deviation of VRP at high VIX (14.3% versus 4.2%) indicates that the VRP becomes noisier during crises, which may have contributed to erroneous conclusions about sign flips based on short sub-samples (see also Bekaert et al., 2022, for discussion of time-varying risk appetite).

The VRP compression is itself a manifestation of absorption. When VIX is high, option writers are pricing in a large amount of expected volatility. Realized volatility is also high, but VIX "over-predicts" by a smaller margin, reflecting the fact that the market's fear has already absorbed much of the tail-risk premium. The insurance premium is lower not because risk has decreased, but because the price of insurance has already been marked up to incorporate the crisis-level risk environment.

6 Economic Implications

The volatility absorption effect has direct and economically significant implications for hedging, portfolio rebalancing, and volatility-targeting strategy design. In this section, we translate the statistical findings into actionable guidance for practitioners and discuss the theoretical implications for asset pricing models.

6.1 Hedging Cost-Benefit Analysis

The marginal value of hedging depends on the regime. We define the hedging cost-benefit ratio as the ratio of average loss avoidance (on shock days) to the cost of maintaining a hedge (proxied by the VRP). In calm markets, this ratio is 13.7×: the expected loss on a shock day is nearly 14 times the daily cost of the hedge. In high-fear markets, the ratio drops to 3.6×—still positive, but dramatically lower.

Table 8: Hedging Cost-Benefit Ratio by VIX Regime

VIX Regime	Avg Shock Loss (%)	Daily Hedge Cost (%)	Cost-Benefit Ratio
Calm ($V < 15$)	1.18	0.086	13.7
Elevated ($15 \leq V < 25$)	1.56	0.196	8.0
High ($V \geq 25$)	2.61	0.725	3.6

Notes: Avg shock loss is the mean absolute return on shock days. Daily hedge cost is proxied by the daily VRP ($\text{VIX}^2/252 - \text{RV}/252$). Cost-benefit ratio = shock loss / hedge cost. These are illustrative calculations; actual hedging costs depend on the specific instruments used.

The implication is clear: the marginal value of *incremental* hedging declines sharply as VIX rises. A risk manager who is already hedged in a high-VIX environment gains relatively little from adding more protection. This does not mean that hedging is unnecessary during crises—the absolute shock magnitudes are larger—but rather that the return per dollar of hedging expenditure is lower. Rational hedging should front-load protection during calm markets, when it is cheap and effective, rather than panic-buying protection during crises, when it is expensive and offers diminishing marginal benefit.

6.2 Portfolio Rebalancing

We test whether the marginal value of daily portfolio rebalancing varies across VIX regimes. Using a simple volatility-targeting framework, prior work in our research program (available upon request) finds that the improvement from daily versus monthly rebalancing is primarily driven by the baseline VT mechanism: daily rebalancing achieves a net Sharpe of 1.42, compared to 0.82 for monthly rebalancing, but the tracking error between daily and monthly is only 1.68% annualized with a return correlation of 0.976. The difference narrows substantially when transaction costs are accounted for, and high-VIX regimes do not disproportionately reward more frequent rebalancing.

This is consistent with the absorption framework. If shocks have diminishing marginal impact at high VIX, then the information content of daily VIX updates—which could in principle justify daily rebalancing—also diminishes. The optimal rebalancing frequency is determined by transaction costs rather than by the value of intra-regime information.

6.3 Volatility Targeting Strategy Design

The most direct practical implication concerns the design of volatility targeting (VT) strategies. The standard VT rule sets the equity weight as:

$$w_t = \frac{c}{V_{t-1}} \tag{9}$$

where c is a target volatility parameter (e.g., $c = 12$ for a 12% target) and V_{t-1} is yesterday's VIX level. This rule has a natural connection to absorption:

Proposition 1. *VT as natural absorption handler. A VT strategy with weight $w_t = c/V_{t-1}$ automatically accounts for volatility absorption. When VIX rises from V_0 to $V_1 > V_0$, the strategy reduces exposure by a factor of V_0/V_1, which is consistent with the factor by which the marginal impact of the next shock is reduced (as shown by the absorption regression). The VT weight and the absorption effect are both inversely proportional to VIX, making VT a reasonable heuristic under the absorption hypothesis.*

This observation helps explain the success of simple VT rules documented in the literature (Moreira and Muir, 2017; Harvey et al., 2018). It also suggests why adding complexity to VT—such as momentum overlays, regime-switch indicators, or conditional leverage rules—may fail to improve performance at high VIX levels. Prior work in our research program (available upon request) finds that volatility-of-volatility overlays reduce returns relative to the standard 12/VIX rule (Sharpe 0.53 versus 0.68; DM test $t = -2.81$, $p = 0.005$), and that higher rebalancing frequency yields negligible net improvement once transaction costs are accounted for (daily net Sharpe 1.42 versus monthly 0.82, with the gap driven primarily by the baseline VT mechanism rather than by incremental high-frequency information). These findings are consistent with the absorption framework:

overlays attempt to exploit information in a regime where the marginal information content of fear signals is precisely at its lowest.

6.4 Crisis Response Timing

Absorption has implications for the optimal timing of policy and portfolio responses to crises. If the marginal impact of shocks declines with VIX, then the first shock in a crisis sequence is the most important. Delayed responses—whether by risk managers, central banks, or portfolio managers—face diminishing returns because later shocks carry less marginal information. The "window of opportunity" for effective crisis response is concentrated in the early phase of a volatility spike, before absorption sets in.

Conversely, the end of a crisis (VIX declining from high to calm) should see the marginal impact of positive shocks *increasing*—a symmetry we do not test in this paper but which suggests that the recovery phase deserves more tactical attention than the sustained-crisis phase.

7 Robustness

We subject our main findings to several robustness checks.

7.1 Alternative Shock Thresholds

We re-estimate the absorption regression using $\tau \in \{1, 1.5, 2.5, 3\}$ instead of our baseline $\tau = 2$. Table 9 shows that the absorption coefficient is negative and statistically significant for all thresholds, though the magnitude varies. Lower thresholds ($\tau = 1$) include many small VIX changes and produce a weaker absorption signal ($\hat{\beta} = -0.00015$). Higher thresholds ($\tau = 3$) isolate genuinely large shocks and produce stronger absorption ($\hat{\beta} = -0.00038$), but with fewer observations and wider confidence intervals.

Table 9: Absorption Coefficient Under Alternative Shock Thresholds

Threshold (τ)	N_{shock}	$\hat{\beta}$	t-stat (NW)	p-value
1.0	1,842	-0.00015	-2.31	0.021
1.5	1,287	-0.00022	-2.94	0.003
2.0	893	-0.00028	-3.42	< 0.001
2.5	624	-0.00033	-3.28	0.001
3.0	441	-0.00038	-3.04	0.002

Notes: Dependent variable: $\text{NSI}_t = |r_t^{\text{SPY}}|/V_t$. Independent variable: V_t. Newey-West standard errors with 10 lags. Sample: 2006–2026.

The monotonic increase in $|\hat{\beta}|$ with τ is consistent with the absorption hypothesis: larger shocks are more "absorb-able" because they are more likely to contain information already reflected in a high VIX level.

7.2 Sub-Period Stability

We split the sample into three sub-periods: 2006–2012 (GFC era), 2013–2019 (low-volatility era), and 2020–2026 (COVID and post-COVID era). Table 10 shows that the absorption coefficient is negative in all three sub-periods, though it is strongest during the GFC era ($\hat{\beta} = -0.00035$) and weakest during the low-volatility era ($\hat{\beta} = -0.00018$). The latter is expected: with VIX rarely exceeding 25 during 2013–2017, there is less variation in the ambient fear level to identify the absorption effect.

Table 10: Absorption Coefficient by Sub-Period

Sub-Period	N_{shock}	$\hat{\beta}$	t-stat (NW)	p-value
2006–2012 (GFC era)	378	-0.00035	-2.89	0.004
2013–2019 (Low-vol era)	198	-0.00018	-1.72	0.087
2020–2026 (COVID era)	317	-0.00031	-2.65	0.008

Notes: Same specification as Table 4. Shock threshold $\tau = 2$. Newey-West standard errors with 10 lags.

7.3 Alternative Normalization

Our baseline NSI normalizes absolute returns by the VIX level. An alternative normalization uses the 20-day realized volatility:

$$\text{NSI}_t^{\text{RV}} = \frac{|r_t|}{\sqrt{RV_t^{(20)}}} \tag{10}$$

where $RV_t^{(20)} = \sum_{i=1}^{20} r_{t-i}^2$. We re-estimate the absorption regression using NSI^{RV} as the dependent variable and the 20-day realized volatility as the independent variable. The slope remains negative ($\hat{\beta}^{\text{RV}} = -0.0031$, $t = -2.76$), confirming that the absorption effect is not an artifact of VIX-specific measurement. When we normalize by realized rather than implied volatility, the phenomenon persists.

7.4 Controlling for Return Autocorrelation

VIX changes and SPY returns are contemporaneously correlated: large VIX increases coincide with large SPY declines. This raises the concern that the absorption effect is simply a restatement of the leverage effect—that large negative returns drive both the VIX increase and the denominator of NSI. To address this, we re-estimate the regression controlling for the lagged absolute return:

$$\text{NSI}_t = \alpha + \beta \cdot V_t + \gamma \cdot |r_{t-1}| + \varepsilon_t \tag{11}$$

The absorption coefficient $\hat{\beta}$ remains negative and significant (-0.00025, $t = -3.14$), indicating that the effect is not driven by return persistence.

8 Conclusion

This paper documents and quantifies *volatility absorption*—the diminishing marginal impact of fear shocks on asset prices as the ambient level of market fear rises. Our main findings are:

1. **The absorption effect is real and economically significant.** The shock amplification ratio for SPY declines from 3.16× in calm markets to 2.32× in high-fear markets (VIX 25–30), with a slight reversal to 2.43× in crisis regimes (VIX $\geq$ 30) attributable to extreme outlier events. The normalized regression slope is -0.00028 ($t = -3.42$), and the effect is robust to alternative shock thresholds, normalization methods, and sub-period splits.

2. **Absorption is a cross-asset phenomenon.** Three of four tested assets (SPY, GLD, TLT) exhibit statistically significant negative absorption coefficients. The exception (0050.TW) is consistent with VIX being a US-specific fear gauge.

3. **The endogenous/exogenous distinction is the key mechanism.** Rate shocks and risk-off flights—risks already reflected in VIX—show evidence of absorption (coefficients +0.019 and +0.007). Geopolitical shocks—exogenous to the VIX information set—show no statistically significant absorption (-0.003, $p > 0.10$), suggesting these shocks may retain their impact. This distinction is the paper's central contribution, though the geopolitical shock sample ($N = 89$) limits statistical power.

4. **The VRP narrows but does not flip.** The variance risk premium compresses from +3.5% to +2.8% at high VIX but remains strictly positive. There is no evidence of a VRP sign flip during extreme market stress.

5. **VT strategies naturally handle absorption.** The standard c/VIX weight rule is approximately optimal under the absorption hypothesis. Prior work in our re-

search program (available upon request) suggests that adding overlays at high VIX tends to degrade performance, consistent with this framework.

8.1 Limitations

Several limitations should be noted. First, as discussed in Section 3, the supplementary normalized shock impact measure ($\text{NSI}_t = |r_t|/V_t$) uses VIX in the denominator while also conditioning on VIX level to define regimes. This creates a mechanical negative relationship: as VIX rises, NSI is deflated by a larger denominator, which would produce a declining NSI–VIX pattern even absent a genuine economic absorption effect. For this reason, we designate the SAR—which does *not* divide by VIX—as our primary identification measure. The SAR independently confirms the declining pattern (Table 3), and three additional pieces of evidence argue against a purely mechanical explanation: (i) the alternative normalization using realized volatility (Section 7) produces qualitatively identical results ($\hat{\beta}^{\text{RV}} = -0.0031$, $t = -2.76$); (ii) the endogenous/exogenous decomposition (Table 6) shows differential absorption across shock types that share the same VIX denominator—rate shocks show significant absorption while geopolitical shocks do not—a pattern inconsistent with a purely mechanical VIX-denominator effect; and (iii) within each VIX bin, the SAR controls for the baseline volatility level, isolating the incremental shock effect. Nevertheless, we acknowledge that the NSI regression overstates the economic magnitude of absorption to some degree, and future work using simulation exercises to quantify the mechanical component—for example, using the methodology of Patton (2011)—would further sharpen identification.

Second, our sample covers a single 20-year period. While we show sub-period stability, the universality of the absorption effect across different macroeconomic regimes (e.g., secular inflation versus deflation) remains to be established. Third, our shock type classification is heuristic rather than model-based; a more rigorous identification of endogenous versus exogenous shocks (e.g., using high-frequency data or narrative identification as in Romer and Romer 2004) would strengthen the causal interpretation. Fourth, we use daily data; intraday analysis could reveal whether absorption operates at higher frequencies.

Fifth, our cross-asset sample is limited to four assets; a broader international panel would provide more powerful tests of the VIX-specificity hypothesis. Sixth, the hedging cost-benefit analysis is illustrative rather than based on a fully specified dynamic portfolio optimization. Seventh, the five-bin SAR analysis (Table 3) reveals a non-monotonicity in the crisis regime (VIX ≥ 30), where SAR increases slightly from $2.32\times$ to $2.43\times$. While we attribute this to extreme outlier events and composition effects in crisis periods, it suggests that the absorption effect may weaken or reverse at the most extreme fear levels—a possibility that merits further investigation with larger crisis samples.

8.2 Directions for Future Research

Several extensions are natural. First, testing absorption with local volatility indices (e.g., VIXTWN for Taiwan, V2X for Europe) would determine whether the effect is universal or specific to the VIX/US equity nexus. Second, developing a theoretical model in which absorption emerges endogenously from rational updating—perhaps through a Bayesian learning framework in which agents have diminishing uncertainty updates as they accumulate crisis information—would provide microfoundations. Third, exploring the implications for option pricing—where absorption would imply that out-of-the-money put prices are "too high" during sustained crises relative to their ex-post payoff—could generate testable predictions. Fourth, examining whether the absorption effect extends to non-financial crises (pandemic, climate events) would test the generality of the mechanism. Fifth, the NFP event study could be strengthened substantially by incorporating surprise-magnitude controls following Andersen et al. (2003), using consensus forecast data from Bloomberg or the Federal Reserve Bank of Philadelphia's real-time dataset. Sixth, simulation exercises quantifying the mechanical component of the NSI regression—for example, generating returns under a null of no absorption and measuring the spurious slope attributable to the VIX denominator—would further sharpen identification; the forecast comparison methodology of Patton (2011) offers an alternative approach to this problem.

References

Andersen, T. G., Bollerslev, T., Diebold, F. X., & Vega, C. (2003). Micro effects of macro announcements: Real-time price discovery in foreign exchange. *American Economic Review*, 93(1), 38–62.

Andrei, D., & Hasler, M. (2015). Investor attention and stock market volatility. *Review of Financial Studies*, 28(1), 33–72.

Balduzzi, P., Elton, E. J., & Green, T. C. (2001). Economic news and bond prices: Evidence from the U.S. Treasury market. *Journal of Financial and Quantitative Analysis*, 36(4), 523–543.

Barberis, N., Huang, M., & Santos, T. (2001). Prospect theory and asset prices. *Quarterly Journal of Economics*, 116(1), 1–53.

Baur, D. G., & Lucey, B. M. (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. *Financial Review*, 45(2), 217–229.

Bekaert, G., Engstrom, E. C., & Xu, N. R. (2022). The time variation in risk appetite and uncertainty. *Management Science*, 68(6), 3975–3995.

Bekaert, G., & Wu, G. (2000). Asymmetric volatility and risk in equity markets. *Review of Financial Studies*, 13(1), 1–42.

Bekaert, G., & Hoerova, M. (2014). The VIX, the variance premium and stock market volatility. *Journal of Econometrics*, 183(2), 181–192.

Black, F. (1976). Studies of stock price volatility changes. *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economics Statistics Section*, 177–181.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Bollerslev, T., Tauchen, G., & Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.

Carr, P., & Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3), 1311–1341.

Chernov, M., Lochstoer, L. A., & Lundeby, S. R. (2022). Conditional dynamics and the multi-horizon risk-return trade-off. *Review of Financial Studies*, 35(3), 1310–1347.

Christie, A. A. (1982). The stochastic behavior of common stock variances: Value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.

Da, Z., Engelberg, J., & Gao, P. (2015). The sum of all FEARS: Investor sentiment and asset prices. *Review of Financial Studies*, 28(1), 1–32.

Danielsson, J., Valenzuela, M., & Zer, I. (2018). Learning from history: Volatility and financial crises. *Review of Financial Studies*, 31(7), 2774–2805.

Drechsler, I., & Yaron, A. (2011). What's vol got to do with it. *Review of Financial Studies*, 24(1), 1–45.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

Engle, R. F., & Ng, V. K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778.

Fleming, J., Kirby, C., & Ostdiek, B. (2001). The economic value of volatility timing. *Journal of Finance*, 56(1), 329–352.

Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.

Haas, M., Mittnik, S., & Paolella, M. S. (2004). A new approach to Markov-switching GARCH models. *Journal of Financial Econometrics*, 2(4), 493–530.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.

Harvey, C. R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M., & Van Hemert, O. (2018). The impact of volatility targeting. *Journal of Portfolio Management*, 45(1), 14–33.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.

Lin, C.-C., Chen, C.-S., & Hwang, D.-Y. (2020). Does VIX or volume improve the prediction of intraday returns for the Taiwan stock market? *Pacific-Basin Finance Journal*, 61, 101316.

Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.

Martin, I. (2017). What is the expected return on the market? *Quarterly Journal of Economics*, 132(1), 367–433.

Moreira, A., & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.

Muler, N., & Yohai, V. J. (2008). Robust estimates for GARCH models. *Journal of Statistical Planning and Inference*, 138(10), 2918–2940.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

Romer, C. D., & Romer, D. H. (2004). A new measure of monetary shocks: Derivation and implications. *American Economic Review*, 94(4), 1055–1084.

Todorov, V. (2010). Variance risk-premium dynamics: The role of jumps. *Review of Financial Studies*, 23(1), 345–383.

Vlastakis, N., & Markellos, R. N. (2012). Information demand and stock market volatility. *Journal of Banking & Finance*, 36(6), 1808–1821.

Whaley, R. E. (2000). The investor fear gauge. *Journal of Portfolio Management*, 26(3), 12–17.

Zakoian, J.-M. (1994). Threshold heteroskedastic models. *Journal of Time Series Analysis*, 15(3), 253–266.

A Variable Definitions

Table 11: Variable Definitions

Variable	Definition
r_t	Daily log return: $\ln(P_t/P_{t-1})$
V_t	VIX closing level on day t
ΔV_t	Daily VIX change: $V_t - V_{t-1}$
NSI_t	Normalized shock impact: $\lvert r_t\rvert/V_t$
SAR_j	Shock amplification ratio in regime j: $\overline{\lvert r\rvert}_{S_j}/\overline{\lvert r\rvert}_{N_j}$
VRP_t	Variance risk premium: $\text{VIX}_t^2/252 - RV_t^{(22)}/22$
$RV_t^{(h)}$	Realized variance over h days: $\sum_{i=1}^{h} r_{t-h+i}^2$

B Additional Cross-Asset Results

Table 12 reports the full absorption regression results for each asset, including intercepts and R^2 values.

Table 12: Full Absorption Regression Results by Asset

Asset	$\hat{\alpha}$	$\hat{\beta}$ ($\times 10^4$)	$t(\hat{\beta})$	Adj R^2	N
SPY	0.091	−2.8	−3.42	0.031	893
GLD	0.078	−4.3	−4.17	0.044	893
TLT	0.074	−4.4	−3.89	0.039	893
0050.TW	0.062	+1.9	+1.62	0.008	612

Notes: Dependent variable: $\text{NSI}_t = \lvert r_t\rvert/V_t$. Independent variable: V_t (VIX level). Newey-West standard errors with 10 lags. $\hat{\beta}$ is multiplied by 10^4 for readability. 0050.TW has fewer shock-day observations due to different trading calendar. Sample: 2006–2026.